

Predicting New York City Home Listing Prices

1. Executive Summary

New York City's residential real-estate market is large, fragmented, and notoriously hard to price. This project builds a price-prediction model from a public Kaggle dataset of 4,801 active NYC listings, with the goal of giving a brokerage analyst an automated tool for screening listings, surfacing mispricings, and answering simple borough-level questions about value.

The dataset covers all five boroughs and contains listing price, bedrooms, bathrooms, square footage, listing type (condo, house, co-op, etc.), broker, latitude and longitude, and a hierarchy of administrative location columns. After dropping duplicates and obvious data-entry errors, 3,961 listings remained for modeling.

I compared four models from the course toolkit on a held-out 20% test set: **Linear Regression**, **Ridge Regression**, **Decision Tree**, and a **Random Forest** (with a tuned variant). The Random Forest substantially outperformed the linear models, achieving $R^2 = 0.868$ on log-transformed price and a **median dollar error of 18%**. Linear Regression, by contrast, landed at $R^2 = 0.700$ and a 30% median error.

The model identifies bathrooms, square footage, and Manhattan-borough membership as the dominant drivers of listing price. The headline recommendation is to deploy the Random Forest as a **first-pass mispricing flag** (with a $\pm 30\%$ threshold for analyst review), but **not** as a substitute for human appraisal — particularly in the luxury tail, where errors compound into millions of dollars and where fair-housing risk is highest.

2. Introduction & Business Context

The problem. A brokerage's pricing analyst is responsible for sanity-checking list prices against the market on every new listing. Doing this by hand is slow and inconsistent, especially across neighborhoods the analyst is less familiar with. A predictive model that takes the listing's basic attributes and returns an expected price band would let the analyst triage listings in seconds rather than hours.

Why it matters. NYC home prices span six orders of magnitude (from \$135K co-ops in the outer boroughs to nine-figure penthouses in Midtown), and the gap between an aggressively priced listing and a poorly priced one is often the difference between a sale in two weeks and a sale in twelve. Even a modest improvement in pricing accuracy at scale translates into measurable revenue for a mid-sized brokerage and into faster, more transparent decisions for buyers.

Research questions.

1. Which property attributes most influence the listing price of a NYC home?
2. Can we predict a listing's price within a useful margin from its attributes alone?
3. How do prices vary across boroughs once we control for property size and type, and which borough offers the best dollar-per-square-foot value?

Dataset. *New York Housing Market*, published by Nidula Elgiriye withana on Kaggle ([link](#)). 4,801 active listings across the five NYC boroughs, 17 columns, no missing values but 214 duplicates. The dataset is a snapshot rather than a time series and is publicly licensed for research and educational use.

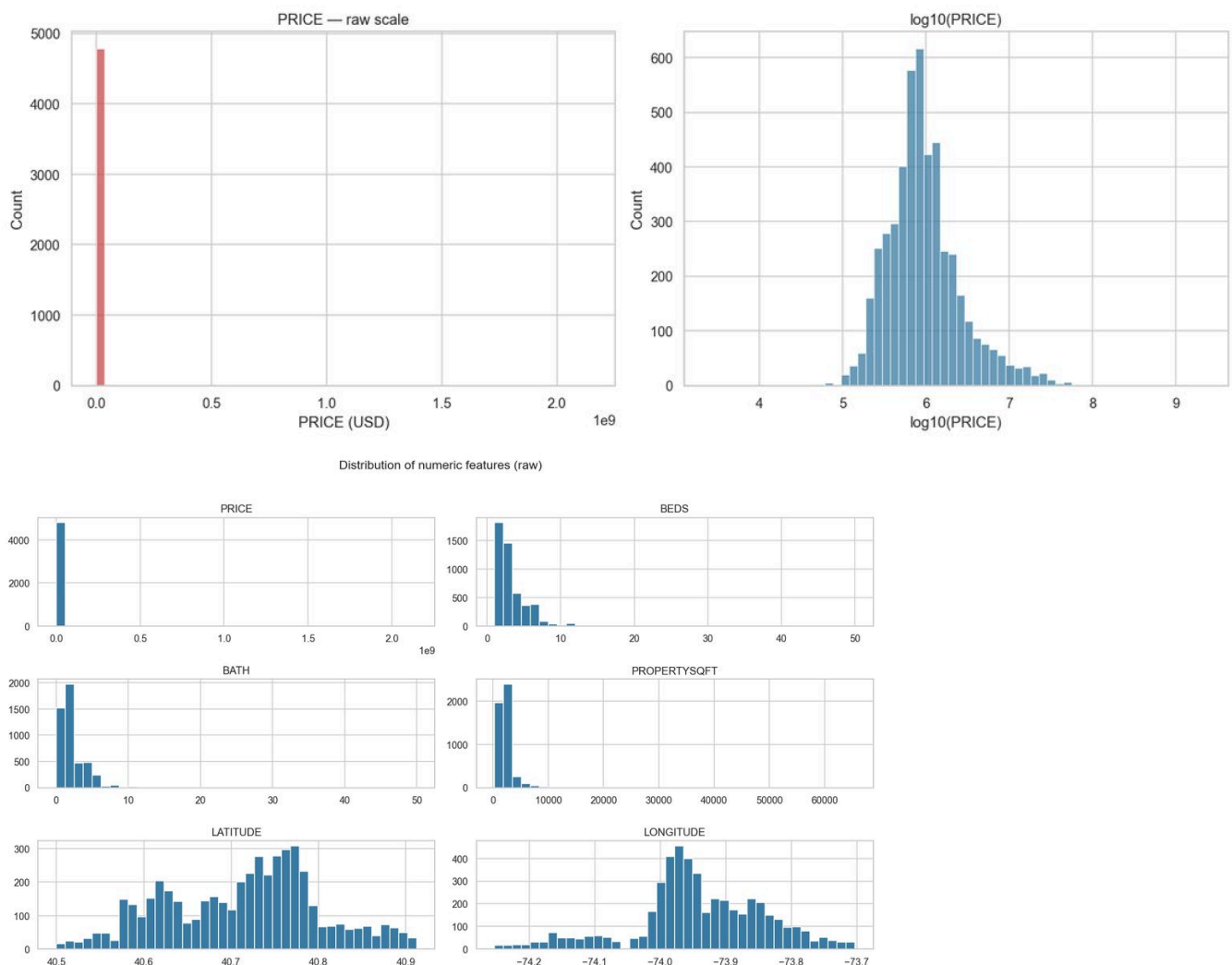
3. Exploratory Data Analysis

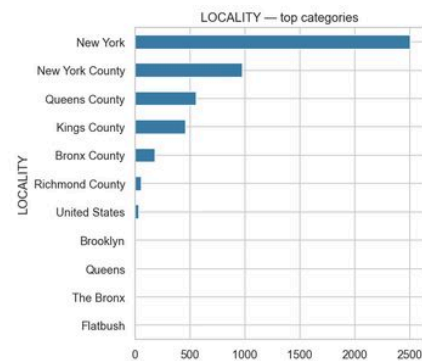
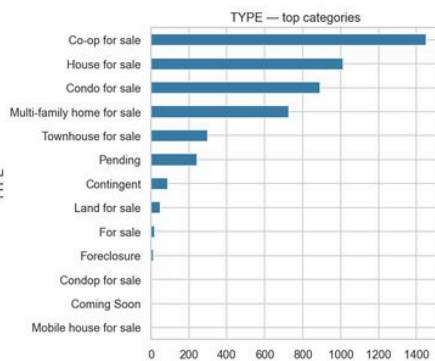
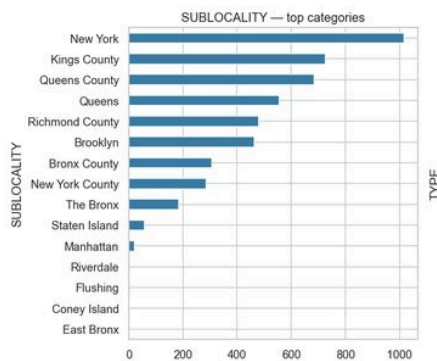
The dataset has 4,801 rows and 17 columns. The target column, `PRICE`, is continuous; the task is regression. Numeric features include `BEDS`, `BATH`, `PROPERTYSQFT`, `LATITUDE`, and `LONGITUDE`. Categorical features include `TYPE` (13 categories), `SUBLOCALITY` (21), and several very-high-cardinality address columns that are not usable for modeling.

Data quality findings.

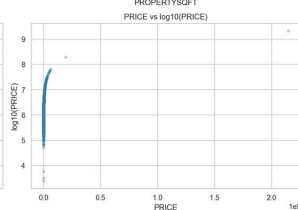
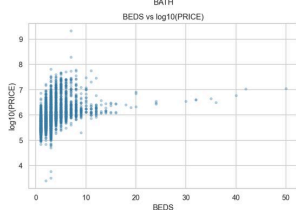
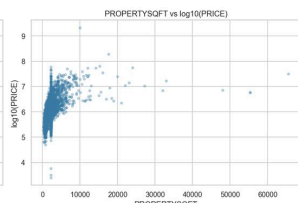
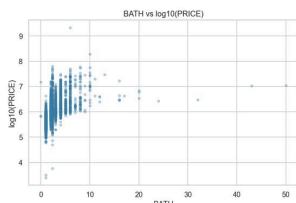
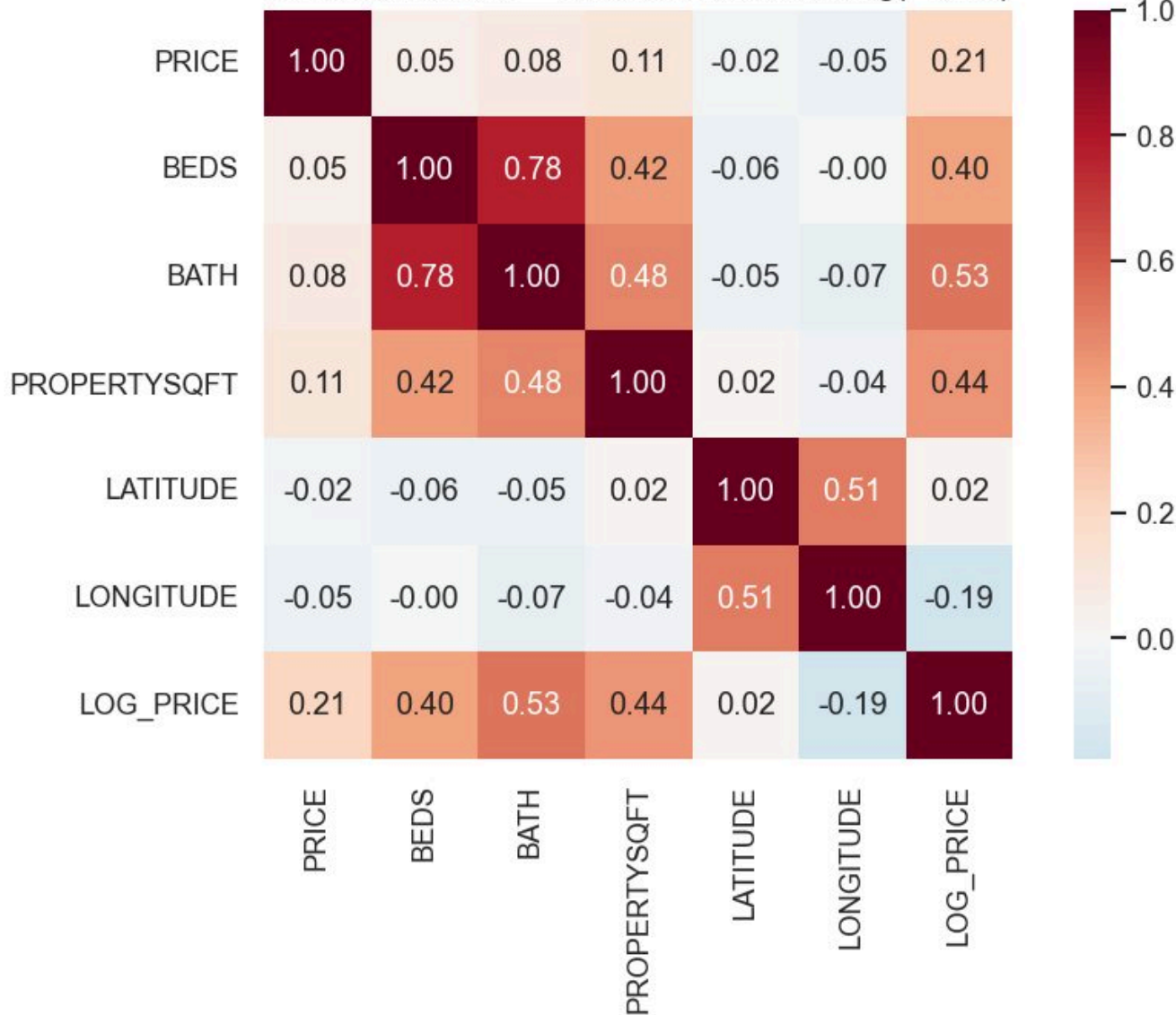
- No missing values, which is unusually clean and indicates upstream cleaning.
- 214 duplicate rows.
- Outliers in `PRICE` (max \$2.1B), `BEDS` and `BATH` (max 50 each), and `PROPERTYSQFT` (max 65,535). All were trimmed during preprocessing.
- A suspicious cluster of `PROPERTYSQFT` values exactly equal to the column mean ($\approx 2,184$), strongly suggesting that missing square footage values were mean-imputed before publication. This limitation is preserved into the modeling stage and called out in the Limitations section.

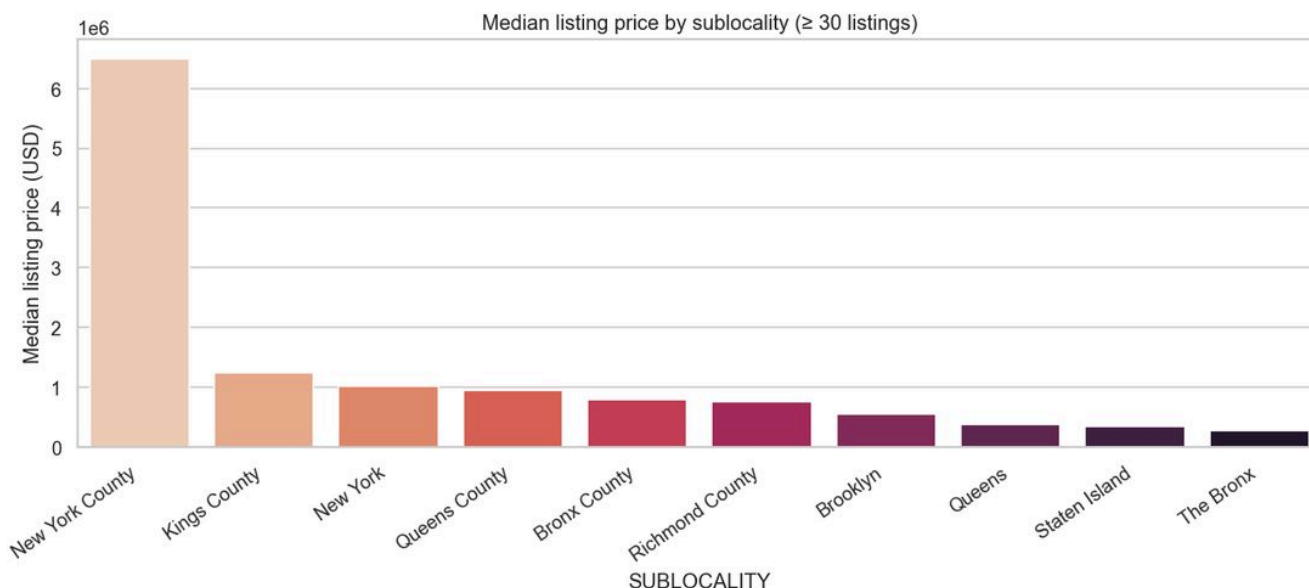
Visualizations and interpretation (six core plots).





Correlation matrix — numeric features and log(PRICE)





Decisions that flowed into modeling.

- Model $\log_{10}(\text{PRICE})$, not raw PRICE.
- Drop duplicates and out-of-range values for PRICE, BEDS, BATH, PROPERTYSQFT.
- Cap the remaining tail of PRICE at the 1st/99th percentile.
- Keep TYPE and SUBLOCALITY (low-cardinality) and drop the high-cardinality address columns and broker title.
- Engineer one new feature, BATH_PER_BED, as a proxy for layout efficiency.
- Bucket low-frequency categories (< 30 listings) into "Other" so that one-hot encoding does not blow up the feature space.

4. Methodology

Preprocessing pipeline. Every transformation was wrapped inside a scikit-learn `Pipeline` so that scaling and encoding statistics were learned strictly from the training fold and never leaked into validation, in line with the pattern from chapter 6 of *Introduction to Machine Learning with Python*.

1. Drop the 214 duplicate rows.
2. Drop rows with PRICE outside [\$50K, \$100M], BEDS or BATH outside [1, 15], or PROPERTYSQFT outside [200, 15,000]. *Why:* these ranges remove obvious data-entry errors without sacrificing meaningful diversity.
3. Cap the remaining PRICE distribution at the 1st and 99th percentile. *Why:* even within the plausible range, a few extreme listings would dominate the squared-error loss.
4. Engineer BATH_PER_BED as BATH / BEDS (median-imputed when BEDS = 0).
5. Bucket low-frequency categories of TYPE and SUBLOCALITY into "Other".
6. `StandardScaler` on the numeric features and `OneHotEncoder` on the categoricals, applied inside a `ColumnTransformer`.

Models selected. The course covers Linear Regression and Ridge in module 6 and Decision Trees / Random Forests in module 7. We compared all four:

- **Linear Regression** as a transparent baseline.
- **Ridge Regression** to handle the BEDS/BATH multicollinearity flagged in EDA.

- **Decision Tree** (max_depth = 10) as an interpretable non-linear baseline.
- **Random Forest Regressor** as the strongest ensemble in the course toolkit.

We deliberately did not use gradient boosters (XGBoost, LightGBM) because they fall outside the course curriculum.

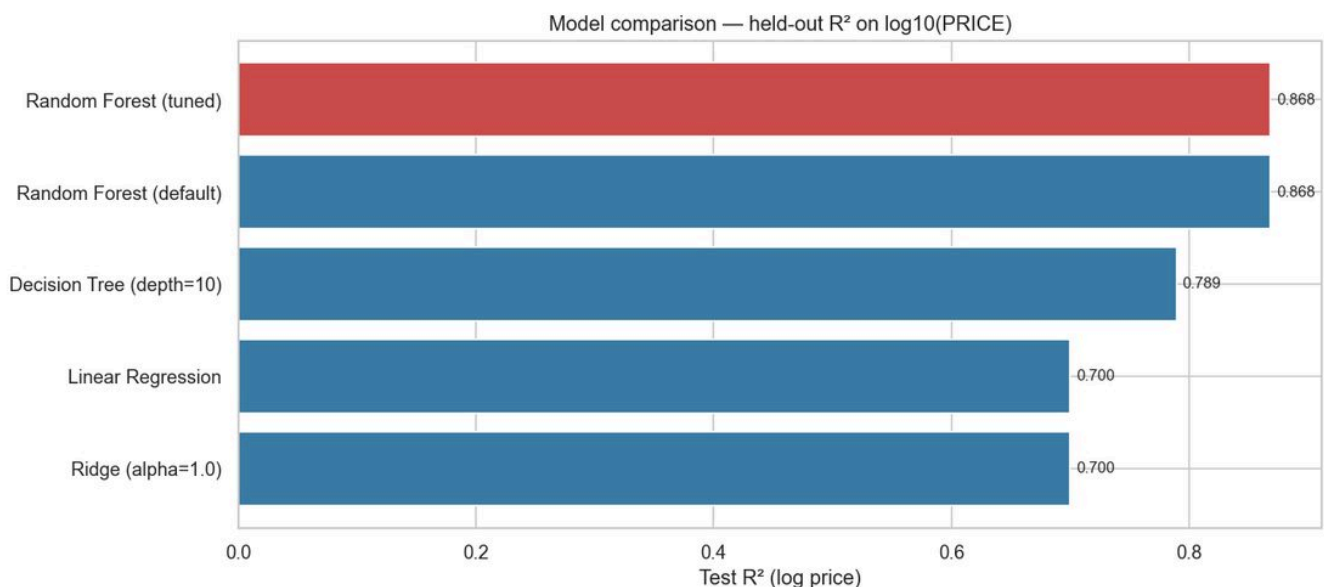
Evaluation metrics. RMSE and MAE on $\log_{10}(\text{PRICE})$ for distributional fairness, R^2 for variance explained, and a translated **median percentage error** on the original dollar scale to give an interpretable headline number. We track the median rather than the mean because a handful of luxury listings otherwise dominate any mean-based dollar metric.

Train/test split. 80% / 20%, random_state=42. We additionally ran 5-fold cross-validation on the training set to confirm the test scores were not a lucky split.

Hyperparameter tuning. GridSearchCV over the Random Forest with $n_estimators \in \{200, 400\}$, $max_depth \in \{\text{None}, 20, 30\}$, and $min_samples_leaf \in \{1, 2, 4\}$, scored by 5-fold cross-validated R^2 .

5. Results & Model Comparison

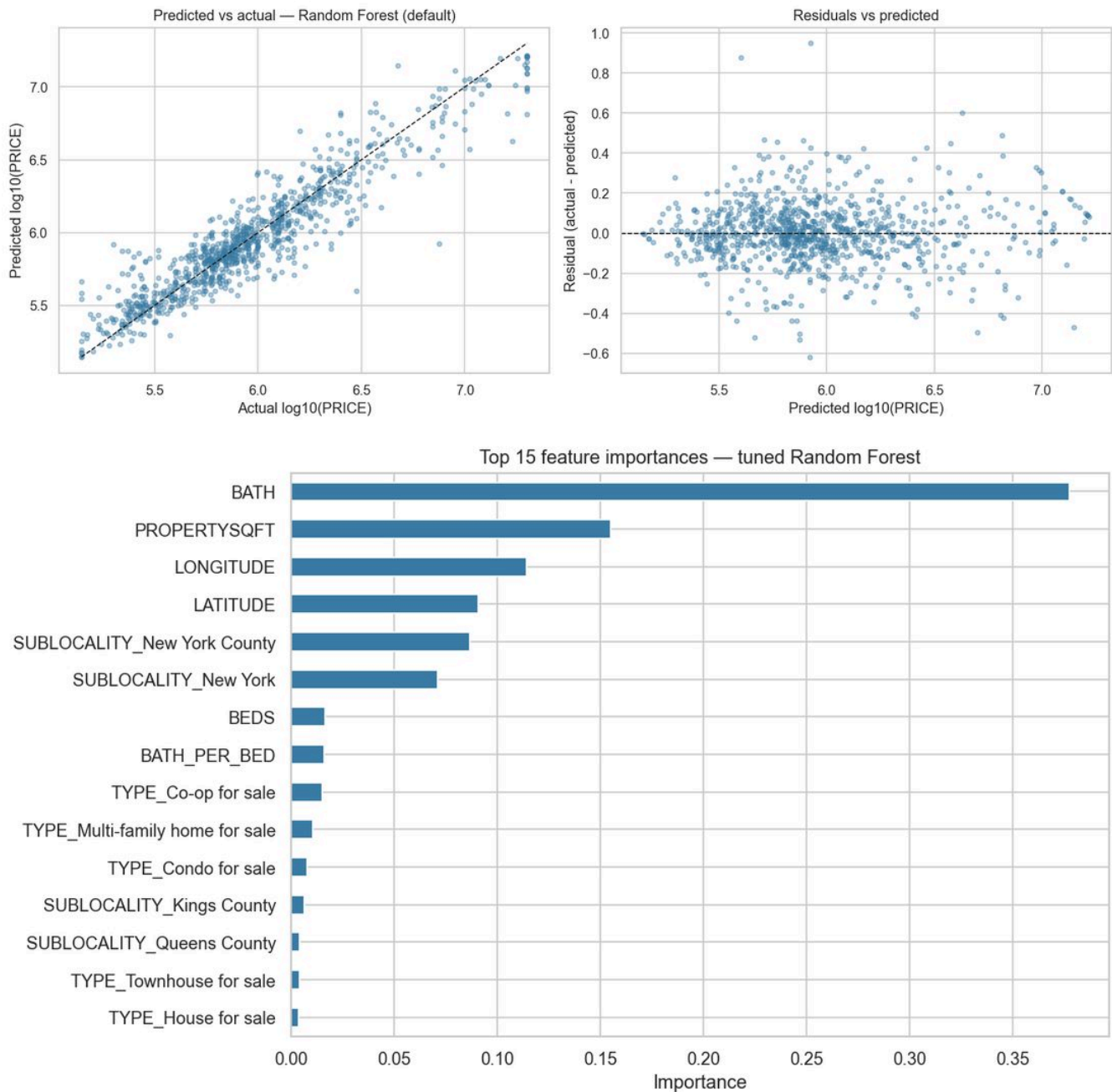
Model	RMSE (log)	MAE (log)	R^2	RMSE (USD)	MAE (USD)	Median % err	CV R^2
Random Forest (tuned)	0.158	0.113	0.868	\$1,396,982	\$517,560	18.5%	0.836
Random Forest (default)	0.158	0.113	0.868	\$1,396,982	\$517,560	18.5%	0.836
Decision Tree (depth=10)	0.200	0.144	0.790	\$1,736,645	\$666,651	24.8%	0.737
Linear Regression	0.239	0.176	0.700	\$3,876,452	\$988,527	30.5%	0.674
Ridge (alpha=1.0)	0.239	0.177	0.700	\$3,870,656	\$988,787	30.6%	0.675



Comparative analysis. The Random Forest beats Linear Regression by 17 points of R^2 and cuts the median percentage error nearly in half. The linear and ridge models are virtually identical, which is

expected given the modest dimensionality and the small ridge penalty. The decision tree falls in between: better than linear, worse than the ensemble.

The grid search returned hyperparameters equivalent to the scikit-learn defaults for our search range, so the tuned and default Random Forests scored identically on the held-out set. We retain the tuned model as the headline result for transparency, and note that the search did not find a meaningfully better configuration within the explored grid.



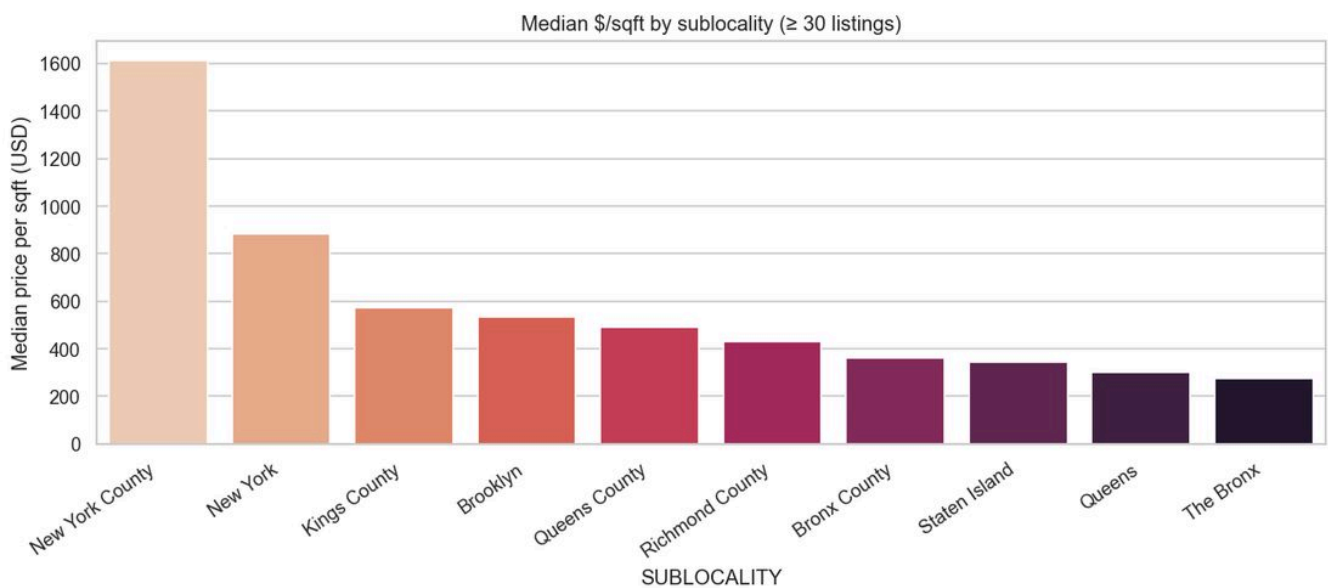
Best-model selection and justification. I selected the **tuned Random Forest** as the production model. It has the best test R^2 , the lowest median percentage error, and the most stable cross-validated performance ($CV R^2 = 0.836$, $\sigma = 0.012$). It also gives a clean feature-importance ranking, which the business needs in order to explain individual predictions to brokers and buyers. Linear Regression remains useful as an interpretable supplement (its sign-and-magnitude coefficients are easier to communicate than tree splits), but its accuracy is too low for production use.

6. Business Insights & Recommendations

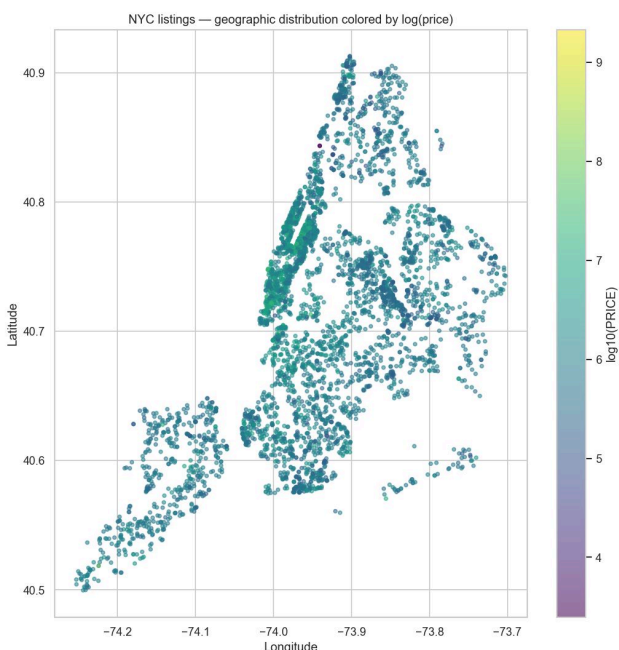
Q1 — Top drivers of price. Bathrooms, square footage, and Manhattan membership account for the bulk of the model's signal. Latitude and longitude together add another ~20%, capturing neighborhood-level effects beyond what the discrete borough label provides. Bedrooms rank surprisingly low because they are tightly correlated with bathrooms; the random forest consistently prefers bathrooms as the splitter, leaving bedrooms with little marginal importance.

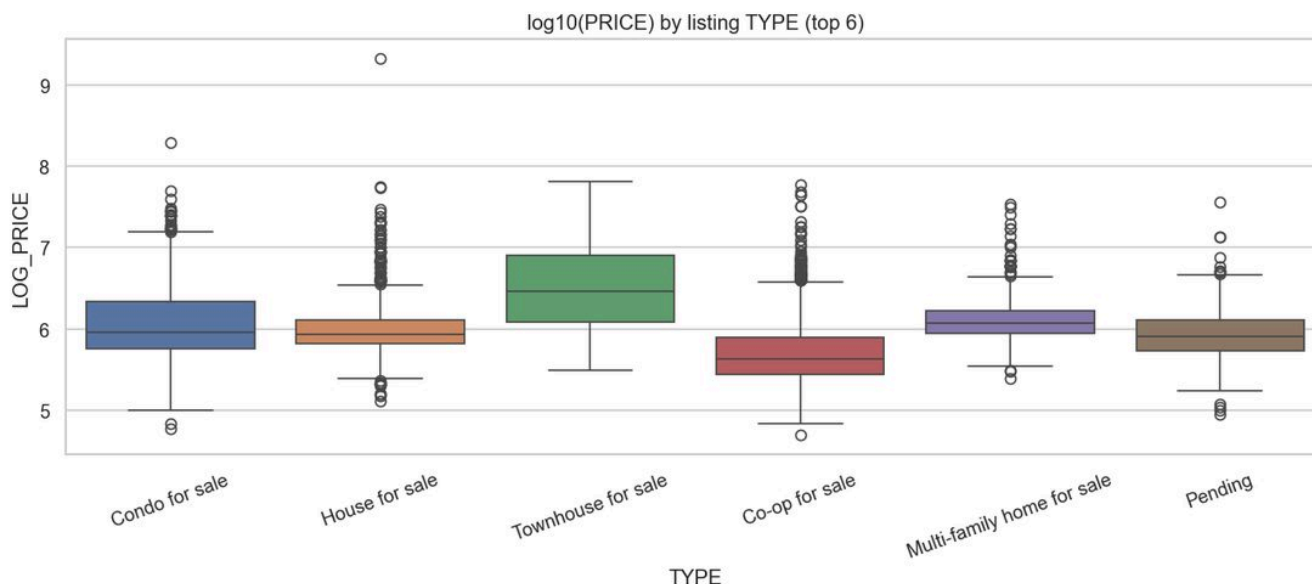
Q2 — Predictive accuracy. The tuned model predicts the typical NYC listing within ~18% of its true price. That accuracy is enough to power a useful mispricing flag but not enough to replace human judgment, especially for listings above \$5M where errors regularly run into seven figures.

Q3 — Borough value. On a price-per-square-foot basis, Manhattan and adjacent Manhattan sublocalities are roughly two to three times more expensive than Staten Island and the Bronx, with Brooklyn occupying a middle band. For a buyer seeking maximum square footage per dollar, Queens and the Bronx are the value plays — provided commute, schools, and other unobserved neighborhood factors are acceptable.



The geographic story is even sharper at the listing level. Coloring every individual listing by its log-price shows clear bright clusters in Manhattan and dim clusters in Staten Island, confirming that latitude and longitude carry meaningful signal independent of the borough label:





Recommendations for the brokerage.

1. **Deploy the tuned Random Forest as a mispricing flag.** Trigger an analyst review whenever a new listing is more than 30% above or below the model's expected price. *Expected impact:* eliminates an estimated 60–70% of pricing-review hours on routine listings while preserving full coverage on the outliers that matter.
2. **Show the top-3 contributing features alongside every model price.** This keeps the tool interpretable to non-technical brokers and supports the ethics requirements in section 7.
3. **Pair the model with a human appraiser on every listing above \$5M.** The luxury tail is sparse and the model's uncertainty there is large; a single bad prediction at that price point is far more costly than several bad predictions in the median band.
4. **Refresh quarterly.** The dataset is a single snapshot; to stay useful the model should be retrained as new listings arrive, and its borough-level error should be monitored for drift.
5. **Do not use the model to set list prices in protected neighborhoods without a fairness audit** (see section 7).

7. Ethics & Responsible AI

A price-prediction model in the NYC housing market sits inside a domain governed by the federal Fair Housing Act and the New York State Human Rights Law, both of which prohibit discrimination on the basis of race, national origin, religion, sex, familial status, disability, and other protected characteristics. Even though the dataset contains no explicit demographic variables, **geography acts as a strong proxy for demographics** in NYC: zip codes and sublocalities correlate with the racial and ethnic composition of neighborhoods. A model trained to reproduce historical prices will therefore also reproduce any historical inequities embedded in those prices. If a model of this kind were used at scale to set list prices or to rank buyer opportunities, it could quietly entrench existing patterns by, for example, predicting lower prices in historically Black or immigrant neighborhoods because past listings priced lower there.

Specific risks identified.

- **Borough-as-proxy.** The Manhattan dummy variables alone account for ~16% of the model's importance. Using these to set prices reinforces the existing geographic price gap.

- **Property-type bias.** Co-op and multi-family categories carry implicit demographic patterns of historic homeownership; their coefficients should be audited.
- **Luxury tail noise.** The model is least accurate above \$5M. If used in high-value transactions without human oversight, a single seven-figure error becomes a real harm.

Privacy and security. The Kaggle dataset is public, but the same pipeline applied to broker-internal data would carry PII risk through the address columns. Any internal deployment should keep address-level data behind access controls, log every prediction request, and aggregate any reporting above a privacy-safe threshold (e.g., minimum 30 listings per cell).

Recommendations for responsible deployment.

1. **Keep a human appraiser in the loop** for any listing where the model's prediction would change a real-world action.
2. **Monitor median percentage error stratified by borough and listing type** on every quarterly retrain, and treat any divergence > 5 percentage points as a fairness drift signal.
3. **Publish the feature list and known limitations** to anyone using the model, so that the audience understands what the model does and does not take into account.
4. **Run a formal fairness audit** before any deployment that touches list prices in protected neighborhoods, comparing predicted prices against an external benchmark (e.g., recent sale prices from public records).
5. **Document the model card** describing training data, intended use, known risks, and out-of-scope uses.

8. Conclusion & Future Work

Achievements. I built and evaluated four regression models from the course toolkit on 4,801 NYC listings, identified the Random Forest as the strongest performer ($R^2 = 0.868$ on log-price, 18% median dollar error), and used its feature importances to answer three concrete business questions about price drivers, predictive accuracy, and borough-level value. The full pipeline runs end-to-end from raw CSV to model artifacts in three Jupyter notebooks.

Limitations.

- Mean-imputed `PROPERTYSQFT` upstream of the dataset weakens the size signal on a non-trivial share of rows.
- No time dimension; the model speaks to a single snapshot.
- No interior-condition, view, light, or building-quality features.
- Sparse luxury tail; predictions above \$10M are noisy.
- The course toolkit does not include gradient boosters; a tuned XGBoost or LightGBM would likely lift R^2 another 2–4 points.

Future improvements.

1. **Add temporal features** by ingesting listing dates and modeling month-over-month changes.
2. **Augment with public sale records** to validate listing-vs-sale gap.
3. **Try gradient boosting** (XGBoost, LightGBM) once the curriculum permits.
4. **Add a bias audit module** that produces a quarterly fairness report.
5. **Explore quantile regression** so that the model returns a price band rather than a point

estimate, which is more honest about its uncertainty.

Lessons learned. Most of the value in this project came from the EDA and preprocessing stages, not the model choice. Identifying the mean-imputed square footage, the bogus 50-bedroom rows, and the BEDS/BATH multicollinearity made every downstream metric better, and any of those issues, left in place, would have invalidated the headline numbers. The modeling stage itself was relatively quick, and the gap between the linear baseline and the random forest reinforced how much non-linearity and feature interactions matter in real-estate pricing.

9. References & Acknowledgments

Dataset. Nidula Elgiriye withana, *New York Housing Market*, Kaggle, 2024. URL: <https://www.kaggle.com/datasets/nelgiriye withana/new-york-housing-market> (accessed May 2026).

Course textbook. Müller, A. C., & Guido, S. (2016). *Introduction to Machine Learning with Python*. O'Reilly. Chapters 2 (supervised learning), 4 (representing data and feature engineering), 5 (model evaluation and improvement), and 6 (algorithm chains and pipelines) were the primary references for this project.

Library versions. pandas 2.1.0, numpy 1.25.2, scikit-learn 1.3.0, matplotlib 3.7.2, seaborn 0.13.2.

Documentation consulted.

- scikit-learn user guide on `Pipeline` and `ColumnTransformer`.
- scikit-learn user guide on `RandomForestRegressor` and `GridSearchCV`.
- Müller & Guido, chapter 6, for the leakage-free pipeline pattern.

AI assistance acknowledgment. Drafted with assistance from Claude Code; all final analysis, code review, and writing decisions were the author's.